



Whitepaper v2.0 • Green Layer-1 Protocol

VERDIS CHAIN

The world's first carbon-negative Layer-1 blockchain.
Built with Substrate, powered by native DPoS
consensus, AMM DEX, ink! smart contracts, and on-
chain carbon credit tracking.

100B

TOTAL
SUPPLY

DPoS

CONSENSUS

30+

PALLETS

\$500M

FDV

CONTENTS

Table of Contents

- 1.** Executive Summary
- 2.** Problem Statement
- 3.** Solution Overview
- 4.** Technical Architecture
- 5.** DPoS Consensus
- 6.** AMM DEX
- 7.** Carbon Credits & Eco Layer
- 8.** EvolvixOS Integration
- 9.** Tokenomics & Distribution
- 10.** Vesting & Fundraising
- 11.** Roadmap
- 12.** Future Solutions
- 13.** Team
- 14.** Conclusion

SECTION 1

Executive Summary

Verdis Chain is a carbon-negative Layer-1 blockchain built on Substrate, powered by native Delegated Proof-of-Stake (DPoS) consensus, an integrated AMM decentralized exchange (DEX), ink! smart contracts, and on-chain carbon credit tracking. The protocol is designed to prove that blockchain infrastructure can be environmentally regenerative rather than destructive.

The native token, **VRDX**, has a fixed supply of **100 billion** with **9 decimals**, serving as the gas token, staking asset, governance instrument, and medium of exchange across the ecosystem. VRDX is not an ERC-20 wrapper — it is woven into the consensus layer itself.

Verdis Chain connects to **EvolvixOS**, an AI Engineering Operating System that provides smart contract auditing, AI-powered development tools, and a plugin marketplace. Together, they form a complete green technology stack: blockchain provides trust and value transfer, AI provides intelligence and automation.

The protocol features **30+ custom Substrate pallets**, including DPoS consensus with 21 target validators, an AMM DEX with 6 liquidity pools, carbon credit minting and retirement, green validator scoring, reforestation logging, governance, IBC cross-chain communication, and Solana-inspired innovations (Gulf Stream, Turbine, Sealevel execution, ZK compression).

Key Metrics

100B TOTAL SUPPLY	6s BLOCK TIME	21 VALIDATORS	6 DEX POOLS
30+ PALLETES	\$500M FDV	\$18M RAISED	8B TGE CIRC.

SECTION 2

Problem Statement

The Environmental Cost of Blockchain

Traditional blockchain networks consume enormous amounts of energy. Bitcoin's annual energy consumption rivals entire countries, and Ethereum's pre-merge PoW system had a carbon footprint comparable to a medium-sized nation. This environmental cost has made blockchain technology a target for criticism from environmentalists, regulators, and institutions.

The Fragmentation Problem

Carbon credit markets are fragmented across multiple registries (Verra, Gold Standard, EU ETS) with inconsistent standards, opaque verification processes, and limited interoperability. There is no unified, transparent infrastructure for issuing, trading, and retiring carbon credits with full chain of custody.

The AI-Blockchain Gap

AI and blockchain have developed as separate ecosystems with limited integration. Smart contract auditing is manual and expensive. There is no protocol where AI models automatically analyze, secure, and optimize on-chain activity in real time.

SECTION 3

Solution Overview

Verdis Chain addresses these problems with a comprehensive green blockchain ecosystem:

Carbon-Negative Consensus

DPoS consensus uses 99.9% less energy than PoW. Validators are incentivized to use renewable energy through green validator scoring, which rewards eco-friendly operators with higher block production priority.

On-Chain Carbon Credits

Carbon credits are minted, traded, and retired directly on-chain with full transparency. Every tonne of CO2 offset is traceable from source to retirement, eliminating double-counting and fraud.

AI-Powered Security

Every smart contract deployed on Verdis Chain is automatically analyzed by EvolvixOS AI before execution, with security scores and vulnerability reports published on-chain.

Native AMM DEX

A built-in automated market maker provides decentralized trading without external dependencies. 6 liquidity pools with overflow-protected arithmetic and deadline parameters.

IBC Cross-Chain

Inter-Blockchain Communication protocol enables trustless bridges to Polkadot, Cosmos, Ethereum, and BSC. Carbon credits become tradeable across all connected chains.

SECTION 4

Technical Architecture

PARAMETER	VALUE	DESCRIPTION
Framework	Substrate	Modular blockchain framework by Parity
Consensus	DPoS + BABE/GRANDPA	Block production + finality gadget
Block Time	6 seconds	BABE slot time target
Smart Contracts	ink! / WASM	Substrate-native contracts via pallet_contracts
Token	VRDX (9 decimals)	Native gas + staking + governance
Supply	100,000,000,000	Fixed supply, disinflationary emission
DEX	Native AMM	6 pools, constant product formula
Cross-Chain	IBC	Inter-Blockchain Communication
Validator Target	21	DPoS active validator set
Pallets	30+	Custom Substrate pallets

Core Pallets

PALLET	INDEX	FUNCTION
dpos	10	Delegated Proof-of-Stake consensus
amm-dex	20	Automated market maker DEX
eco	30	Carbon credits, green scoring, reforestation
fungible-tokens	40	Custom token creation
governance	50	Democracy, council, treasury
presale	60	Token presale and vesting
gulf-stream	51	Transaction forwarding (Solana-inspired)
sealevel	55	Parallel execution scheduling
zkc	53	ZK compression

SECTION 5

DPoS Consensus

Verdis Chain uses Delegated Proof-of-Stake (DPoS) combined with BABE/GRANDPA for block production and finality. Token holders delegate their stake to validators who produce blocks on their behalf.

Validator System

- **21 target validators** in the active set, selected by total delegated stake
- **Green validator scoring** rewards validators using 100% renewable energy with higher priority
- **Slashing** penalizes malicious behavior (equivocation, downtime) with configurable severity
- **Registration deposit:** 0 VRDX (subsidized for early validators)
- **Delegation:** Any VRDX holder can delegate to validators

Economic Parameters

PARAMETER	VALUE
Active Validator Count	21
Epoch Duration	50 slots (300s)
Session Period	50 blocks
Staking APR	5-6.67%
Staking Pool	20B VRDX
Min Delegate	1 VRDX

SECTION 6

AMM DEX

The native Automated Market Maker (AMM) DEX is built directly into the protocol via a custom Substrate pallet, providing decentralized trading without external dependencies.

Liquidity Pools

POOL	PAIR	INITIAL LIQUIDITY
Pool 1	VRDX / ECO	500,000 VRDX
Pool 2	VRDX / CARBON	300,000 VRDX
Pool 3	VRDX / TREE	200,000 VRDX
Pool 4	VRDX / GREEN	200,000 VRDX
Pool 5	ECO / CARBON	100,000 ECO
Pool 6	VRDX / REDD	100,000 VRDX

Security Features

- Overflow-protected arithmetic (`checked_mul`) on all swap/liquidity operations
- Mandatory deadline parameters prevent stale transaction execution
- Self-transfer guard prevents flash loan attacks
- Pool bricking mechanism for emergency shutdown
- LP token overflow protection

SECTION 7

Carbon Credits & Eco Layer

The Eco pallet provides on-chain carbon credit infrastructure, making environmental impact verifiable, immutable, and transparent.

Carbon Credit Lifecycle

- **Minting:** Verified reforestation projects issue pCO2 tokens on-chain (admin-gated)
- **Trading:** Credits trade on the native AMM DEX (VRDX/CARBON pool)
- **Retirement:** Credits are permanently burned with immutable proof of offset
- **Verification:** AI-powered satellite imagery + IoT sensor analysis via EvolvixOS

Green Validator Scoring

Validators receive a green score (1-5) based on their energy source. Solar, wind, and hydro validators receive higher scores, translating to higher block production priority. Self-scoring is prevented — only root can update scores.

Reforestation Logging

Reforestation projects are registered on-chain with GPS coordinates, tree count, CO2 sequestration estimates, and progress tracking. Each project can issue carbon credits proportional to verified impact.

SECTION 8

EvolvixOS Integration

EvolvixOS is an AI Engineering Operating System that forms the intelligence layer of the Verdis ecosystem. Together, Verdis Chain and EvolvixOS provide a complete green technology stack.

Integration Points

Smart Contract Auditing

Every contract deployed on Verdis Chain is automatically analyzed by EvolvixOS AI before execution. Security scores and vulnerability reports are published on-chain.

AI-Powered Governance

AI models analyze governance proposals, simulate economic impact, and provide recommendations to VRDX holders.

Carbon Verification

Satellite imagery analysis and IoT sensor data are processed by AI to autonomously verify reforestation progress and carbon sequestration.

Developer Tools

AI-powered code generation, testing, and deployment tools accessible through EvolvixOS.com — the main entry point to the ecosystem.

The Vision

The goal: go from **Idea** → **AI Assistance** → **Application** → **Smart Contract** → **Verdis Blockchain** → **Deployment** → **Ecosystem**, all through a single coherent platform.

SECTION 9

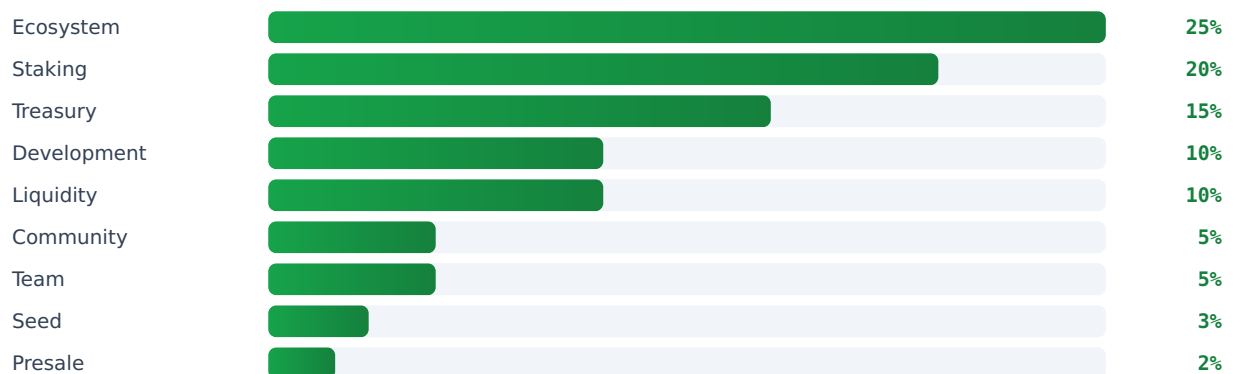
Tokenomics & Distribution

The VRDX token has a fixed supply of 100 billion with 9 decimals, distributed across 9 categories for long-term sustainability.

Allocation

CATEGORY	ALLOCATION	PERCENTAGE	PURPOSE
Ecosystem & Developer Grants	25B	25%	Grants, partnerships, third-party development
PoS Staking Rewards	20B	20%	Validator and delegator block rewards
Treasury	15B	15%	DAO-governed community treasury
Development	10B	10%	Core protocol development
Liquidity	10B	10%	DEX liquidity provisioning
Community	5B	5%	Community rewards, airdrops, faucets
Team & Advisors	5B	5%	Team vesting with 12-month cliff
Seed / Strategic	3B	3%	Early investors, 12-month cliff
Public Presale	2B	2%	Public presale, 6-month cliff

Distribution Chart



SECTION 10

Vesting & Fundraising

Fundraising Rounds

ROUND	PRICE	TOKENS	RAISED	DISCOUNT	CLIFF
Seed	\$0.0015	3B	\$4.5M	70%	12 months
Community	\$0.003	1B	\$3M	40%	3 months
Presale	\$0.004	2B	\$8M	20%	6 months
TGE/IDO	\$0.005	0.5B	\$2.5M	0%	None
Total		6.5B	\$18M		

Vesting Schedule

- **Seed & Private:** 12-month cliff, then linear vesting at 125M VRDX/month
- **Community:** 3-month cliff, then linear vesting
- **Presale:** 6-month cliff, then linear vesting
- **Team & Advisors:** 12-month cliff, then 416.7M VRDX/month over 12 months
- **TGE circulating:** 8B VRDX (8% of total supply)
- **Full unlock:** 10-year schedule (8B → 95B)

Staking Economics

STAKE RATE	APR	ANNUAL REWARDS PER 1000 VRDX
30%	6.67%	66.7 VRDX
35%	5.71%	57.1 VRDX
40%	5.00%	50.0 VRDX

SECTION 11

Roadmap

Phase 1 • Q1 2026

Genesis & TGE

Mainnet genesis launch. 24.5B VRDX initial circulating supply. Team & investor cliff begins.

Phase 2 • Q2 2026

DPOS Staking & DEX Activation

21 active validators. 8.2B VRDX bonded in staking. AMM DEX liquidity bootstrap.

Phase 3 • Q3 2026

Eco Precompiles & Presale Unlock

Carbon offset precompile activation. Presale 3-month cliff completes; linear vesting begins.

Phase 4 • Q1 2027

Seed, Private & Team Cliff End

12-month cliffs end. DAO governance assumes treasury control.

Phase 5 • 2027-2030

Global Carbon Offset Scaling

10,000,000+ tCO2e verified carbon offsets retired globally.

Phase 6 • 2030-2032

AI-Powered Autonomous Governance

EvolvixOS AI analyzes proposals, simulates impact, provides recommendations. Hybrid human-AI governance.

Phase 7 • 2030-2033

Cross-Chain Carbon Credit Protocol

Carbon credits tradeable across all IBC chains. 100M+ tCO2e annual offset capacity.

Phase 8 • 2031-2034

ZK Rollup & 10,000+ TPS

High-throughput microtransactions. Sub-cent fees for green micropayments.

SECTION 12

Future Solutions

Decentralized Identity & Green Certificates

On-chain DID for green energy producers, reforestation projects, and carbon auditors. W3C-compliant verifiable credentials.

Layer-2 ZK Rollup Scaling

10,000+ TPS for carbon credit micro-offsets, IoT data logging, and real-time energy trading with Layer-1 security.

AI-Powered Carbon Verification

Satellite imagery + IoT sensors + AI analysis via EvolvixOS for autonomous reforestation verification.

Global Green Finance

Carbon-backed stablecoins, green bonds, ESG DeFi. Partnership with Verra and Gold Standard. VRDX as global green settlement layer.

IoT & Oracle Network

Decentralized oracle nodes with IoT sensors monitoring air quality, soil health, tree growth, and energy production in real time.

Developer Ecosystem & Grants

5B VRDX grant program for third-party teams building DeFi, NFT, supply chain, and green energy apps on Verdis Chain.

Planetary Carbon Dashboard

Real-time global carbon footprint dashboard. Every tonne of CO2 traceable from source to retirement.

Carbon-Negative Planet (2040)

1B+ tCO2e cumulative offsets. 500+ validators in 50+ countries on 100% renewable energy. Blockchain transitions from environmental liability to solution.

SECTION 13

Team

Dorian Jean**CEO & Founder**

Visionary leader driving the Verdis Chain mission to build the world's first carbon-negative blockchain.

Mark Jamestown**CTO**

Architect of the Substrate-based blockchain infrastructure and consensus design.

Elizabeth Jefferson**Head of Product**

Leads product strategy, user experience, and ecosystem integration across Verdis and EvolvixOS.

Rajs Gordons**Co-Founder & Community**

Founder of Protremix. Leads community growth, marketing, and developer relations.

María Dolores Márquez de Prado**Legal Counsel**

Legal advisor specializing in blockchain regulation, compliance, and international law.

Ignacio Martínez-Arrieta**Legal & Compliance**

Compliance officer ensuring regulatory alignment across jurisdictions and tokenomics.

SECTION 14

Conclusion

Verdis Chain proves that blockchain technology can be environmentally regenerative. By embedding carbon credits, green validator scoring, and reforestation tracking directly into the consensus layer, we make sustainability a protocol-level feature rather than an afterthought.

With 30+ custom pallets, native AMM DEX, ink! smart contracts, DPoS consensus, IBC cross-chain communication, and integration with the EvolvixOS AI ecosystem, Verdis Chain provides a complete platform for the next generation of green decentralized applications.

The 100B VRDX token economy is designed for long-term sustainability — with 45% allocated to ecosystem and staking, 12% to investors with structured vesting, and a DAO-governed treasury ensuring community control. Every economic parameter is designed to incentivize decentralization, security, and environmental impact.

This is not a promise. This is architecture.



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